

EDUCATION

Colorado State University , Fort Collins, CO, USA <i>Doctor of Philosophy, Statistics</i> Thesis: <i>Bayesian Tree Based Methods for Longitudinally Assessed Environmental Mixtures</i> Advisor: Dr. Ander Wilson	2019 – 2024
Colorado State University , Fort Collins, CO, USA <i>Master of Science, Statistics</i>	2019 – 2023
St. Lawrence University , Canton, NY, USA <i>Bachelor of Science, Mathematics & Statistics, Magna Cum Laude</i>	2015 – 2019

EXPERIENCE

Republic of Korea Navy Headquarters , Seoul, South Korea Research Scientist, <i>Naval Future Innovation Research Group</i> Mandatory National Service — discharge scheduled late July 2026	Oct 2024 – Present
Colorado State University , Fort Collins, CO, USA Graduate Research Assistant, <i>Department of Statistics</i> Graduate Teaching Assistant, <i>Department of Statistics</i>	Aug 2019 – Aug 2024 Jun 2021 – Aug 2024 Aug 2019 – May 2021
St. Lawrence University , Canton, NY, USA Student Mentor, <i>Peterson Quantitative Reasoning Center</i> IT Department Student Employee, <i>Information Technology</i>	Aug 2016 – Jun 2019 Aug 2017 – Jun 2019 Aug 2016 – Jun 2019

PUBLICATIONS

Published:

- Leung, M., **Im, S.**, Rowland, S. T., Papatheodorou, S., Coull, B. A., Kioumourtzoglou, M.-A., Weisskopf, M. G., & Wilson, A. (2026). Prenatal exposure to wildfire PM_{2.5} and pregnancy loss in Colorado, USA, 2007–2018. *International Journal of Epidemiology*, 55(1), dyaf212. <https://doi.org/10.1093/ije/dyaf212>
- Im, S.**, Wilson, A., & Mork, D. (2025). Structured Bayesian Regression Tree Models for Estimating Distributed Lag Effects: The R Package dlmtree. *R Journal*, 17(1), 136–159. <https://doi.org/10.32614/RJ-2025-007>
- Mork, D., **Im, S.**, & Wilson, A. (2024). dlmtree: Bayesian Treed Distributed Lag Models. *Comprehensive R Archive Network (CRAN) package*, version 1.1.0.
- Weller, Z. D., **Im, S.**, Palacios, V., Stuchiner, E., & von Fischer, J. C. (2022). Environmental injustices of leaks from urban natural gas distribution systems: Patterns among and within 13 U.S. metro areas. *Environmental Science & Technology*, 56(12), 8599–8609. <https://doi.org/10.1021/acs.est.2c00097>
- 🌟 Selected for American Chemical Society (ACS) Editors' Choice

Manuscripts Under Revision:

- Im, S.**, Mork, D., Leung, M., Weisskopf, M., Kioumourtzoglou, M.-A., & Wilson, A. (2026). Treed Distributed Lag Mixture Model With Zero-Inflated Count Data to Investigate the Association Between Air Pollution and Pregnancy Loss. *Statistics in Medicine*. (Under revision)
- Im, S.**, Mork, D., & Wilson, A. (2026). Multi-Tree Model for Precision Environmental Health Longitudinally Assessed Mixture Exposures. *Journal of the Royal Statistical Society Series C: Applied Statistics*. (Under revision)

SELECTED PROJECTS

Climate Health Impact Assessment 2026

Mar 2026 – Present

Statistical Consultant, Korea Disease Control and Prevention Agency (KDCA)

HONORS & AWARDS

Excellence Award – ROK Army Big Data Analytics Competition	Nov 2025
Excellence Award – ROK Navy & Marine Corps Artificial Intelligence Innovation Challenge	Jun 2025
President, Pi Mu Epsilon (Mathematics Honor Society), St. Lawrence University	Fall 2019
Statistics Dr. O. Kenneth Bates Award, St. Lawrence University	Jun 2019
Dean's list, St. Lawrence University	Fall 2016 – Spring 2019

PRESENTATIONS

The Graduate Student Showcase, Colorado State University, <i>poster</i>	2023
EnviBayes Workshop on Complex Environmental Data, International Society for Bayesian Analysis (ISBA), <i>poster</i>	2023
Joint Statistical Meetings (JSM), American Statistical Association (ASA), Toronto, ON, Canada, <i>oral</i>	2023
The Graduate Student Showcase, Colorado State University, <i>poster</i>	2022
Colorado/Wyoming Chapter, American Association (ASA), <i>oral (virtual)</i>	2021
Joint Statistical Meetings (JSM). American Statistical Association (ASA), Seattle, USA, <i>oral (virtual)</i>	2021
Front Range Student Ecology Symposium (FRSES), Colorado State University, <i>oral (virtual)</i>	2021

TEACHING

Courses

Department of Statistics, Colorado State University

STAT 301: Introduction to Statistical Methods

Spring 2020 – Spring 2021

STAT 201: General Statistics

Fall 2019

Workshops

Department of Statistics, Colorado State University

Coding & Cookies – Python: Working with data

Fall 2023, Spring 2024

Coding & Cookies – Tidy Data in R: Tidyverse

Fall 2021, Fall 2022, Spring 2023

Coding & Cookies – R Basics

Spring 2022

Tutored

Department of Statistics, Colorado State University

STAR 511: Design and Data Analysis for Researchers I – R Software

Spring 2024

Peterson Quantitative Reasoning Center, St. Lawrence University

Fall 2017 – Spring 2019

STAT-325: Probability

STAT-313: Advanced Statistical Models

STAT-213: Applied Regression Analysis

STAT-113: Applied Statistics

CS-140: Intro to Computer Programming

MATH-217: Linear Algebra

MATH-135, 136: Calculus I, II

SKILLS

R: Tidyverse, Rcpp, RStan/brms, R package development, Bayesian modeling

Python: NumPy, SciPy, Pandas, Matplotlib, XGBoost, Scikit-learn, Hyperparameter Optimization

Machine Learning: Gradient Boosted Decision Trees, Random Forest, Ensemble Methods, Feature Engineering

Version Control: Git, GitHub

SQL & High-performance computing (HPC): Proficient

Language: English – Bilingual Proficiency; Korean – Native

Last updated: May 17th, 2026